

MAJOR MOON

Mimas

DISTANCE FROM SATURN 115,208 miles
(185,520 km)

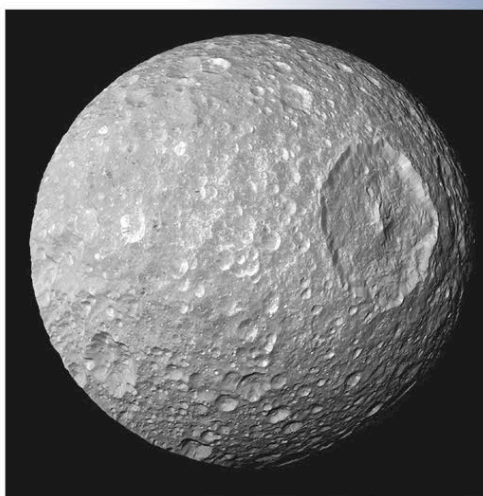
ORBITAL PERIOD 0.94 Earth days

DIAMETER 246 miles (396 km)

Mimas is the first of the major moons out from Saturn, and it orbits the planet in the outer part of the ring system. It is in synchronous rotation, so the same side of the moon always faces the planet in the same way that one side of the Moon always faces Earth. Mimas is a round moon, but it is not a perfect sphere—this icy object is about 19 miles (30 km) longer than it is wide and deep. Its surface is covered in deep, bowl-shaped impact craters. Many of those greater than about 12 miles (20 km) across have central peaks. One crater, Herschel, dwarfs the rest and is the moon's most prominent feature. It is about 80 miles (130 km) wide, almost 6 miles (10 km) deep, and has a prominent central peak. If the impacting body that formed the crater had been much bigger, it might have smashed the moon apart. The crater is named after the astronomer William Herschel, who discovered Mimas on July 18, 1789. It was the sixth of Saturn's moons to be discovered and the first of two discovered by Herschel. Mimas is named after a Titan (see p.190).

**EMPHASIZING DIFFERENCES**

False color highlights slight differences in surface composition on Mimas—for example, the bluish terrain near the crater Herschel, possibly caused by impact ejecta, and greenish terrain elsewhere.

**GIANT CRATER**

The crater Herschel lies on the moon's leading hemisphere (the side that points in the direction in which it is moving) and is about one-third of the diameter of Mimas itself. The impact that formed Herschel must have come close to shattering the moon.

TRUE BLUE

Mimas drifts against the backdrop of Saturn's northern hemisphere in this true-color view. Scattering of sunlight in the relatively cloud-free area gives the planet a bluish hue. The dark lines cutting across the atmosphere are shadows cast by Saturn's rings.

